

SE LAB3 SUBMISSION

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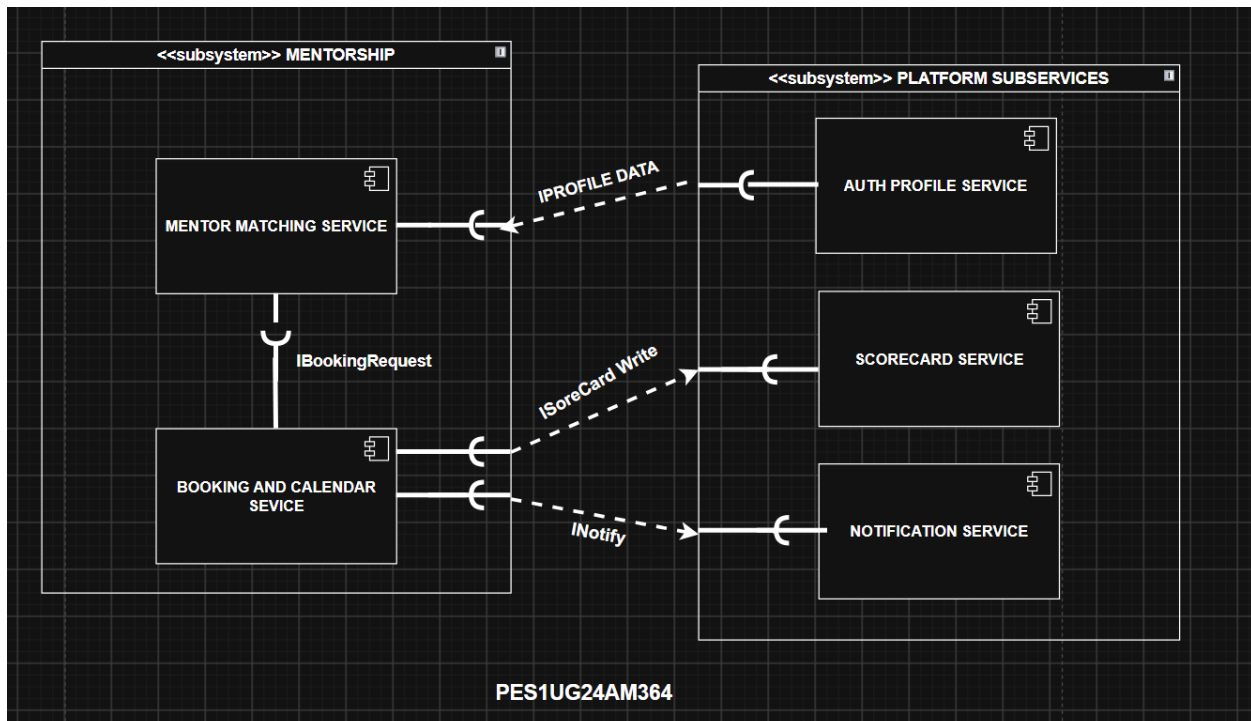
DATE : 4-09-2026

CLASS : AIML 5-G

SCENARIO #6

Deliverables :

1. Component Diagram (PNG)



2. Written Justification

Architecture Selection

We chose Microservices Architecture for the Alumni Mentorship & Mock Interview Platform.

Reason 1: The platform's core functions - mentor matching, calendar booking, and scorecard management - have distinct and independent scaling needs. Matching involves compute-heavy domain-matching logic that may spike during enrollment periods (see NFR-002, which requires ranked results within 2 seconds), while booking and scorecard access have steadier, session-driven load. Microservices let each scale independently without over-provisioning the others.

Reason 2: Fault isolation matters here - if the Scorecard Service goes down, students should still be able to browse mentors and book sessions (FR-001, FR-002, FR-003). A monolithic or tightly layered design would risk a single failure taking down the whole platform.

Security Advantage: Scorecard data is sensitive and subject to soft-deletion within 24 hours of a request (NFR-001). Isolating it as its own service lets it enforce stricter access controls and audit logging independently of less sensitive services like matching or notifications.

Performance Benefit: The Mentor Matching Service can be scaled horizontally during peak demand, such as the start of a semester, without needing to scale the Booking or Notification services, keeping resource usage efficient and helping meet the 2-second response target in NFR-002.